MENDELIO

2023

NEARLY

FINAL MODEL

BREAST CANCER

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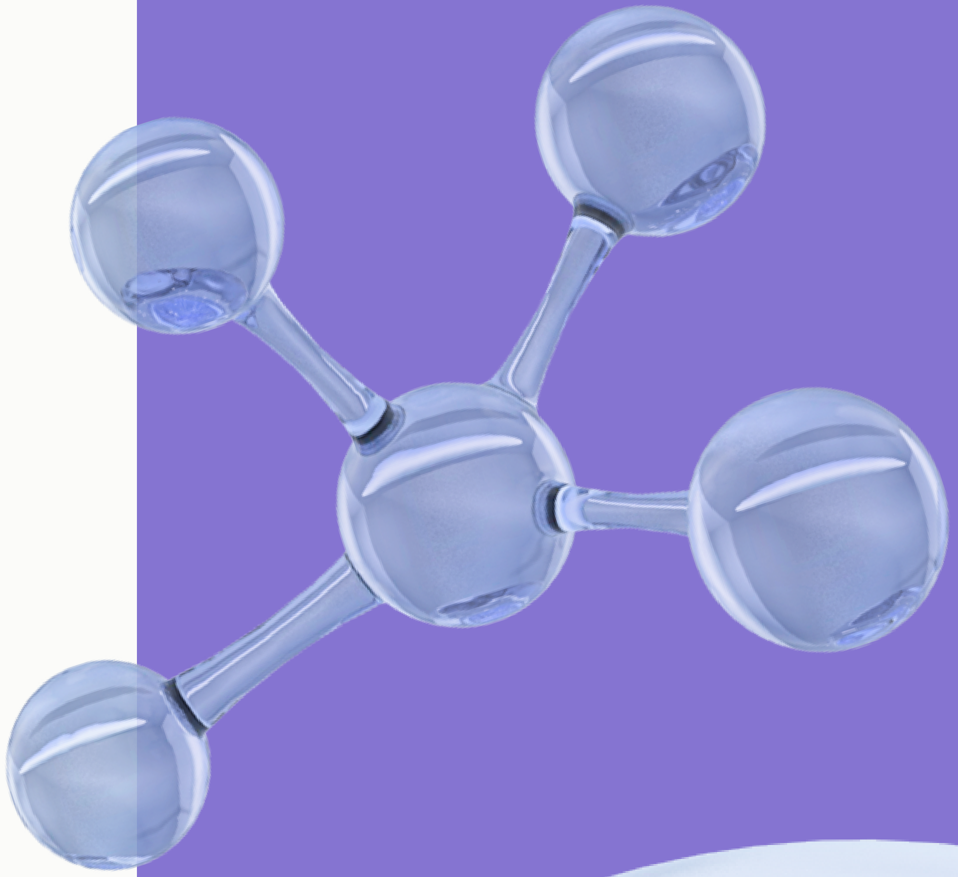
Introduction

In this presentation we will unveil our innovative approach to addressing Breast Cancer, a solution we've carefully developed and consider as our "ultimate" strategy. While it's nearly finalized, we remain open to continual enhancements by adopting successful techniques from other disease management approaches.
Let's explore the key steps that form the foundation of our solution.



In our final experiments, we decided to switch things up by replacing the previous clustering/sub-grouping method. The reason behind this change was that the earlier method didn't show any score improvements, and the recent experiments actually resulted in lower performance.

Instead, we took a different approach by segmenting the kinetic energies of all SNP features into four intervals. We conducted experiments on this new approach using a 10-fold cross-validation, and the results were quite promising. We achieved an approximate AUC of 0.76 and an approximate F1 score of 0.96.

<u>These results make a strong case for adopting this model as the final one for Breast Cancer.</u>

Brief explanations

AUC (Area Under the Curve) and **F1 score** are two commonly used metrics for evaluating the performance of a **PRS** (polygenic risk score). Here's why they are considered good metrics:

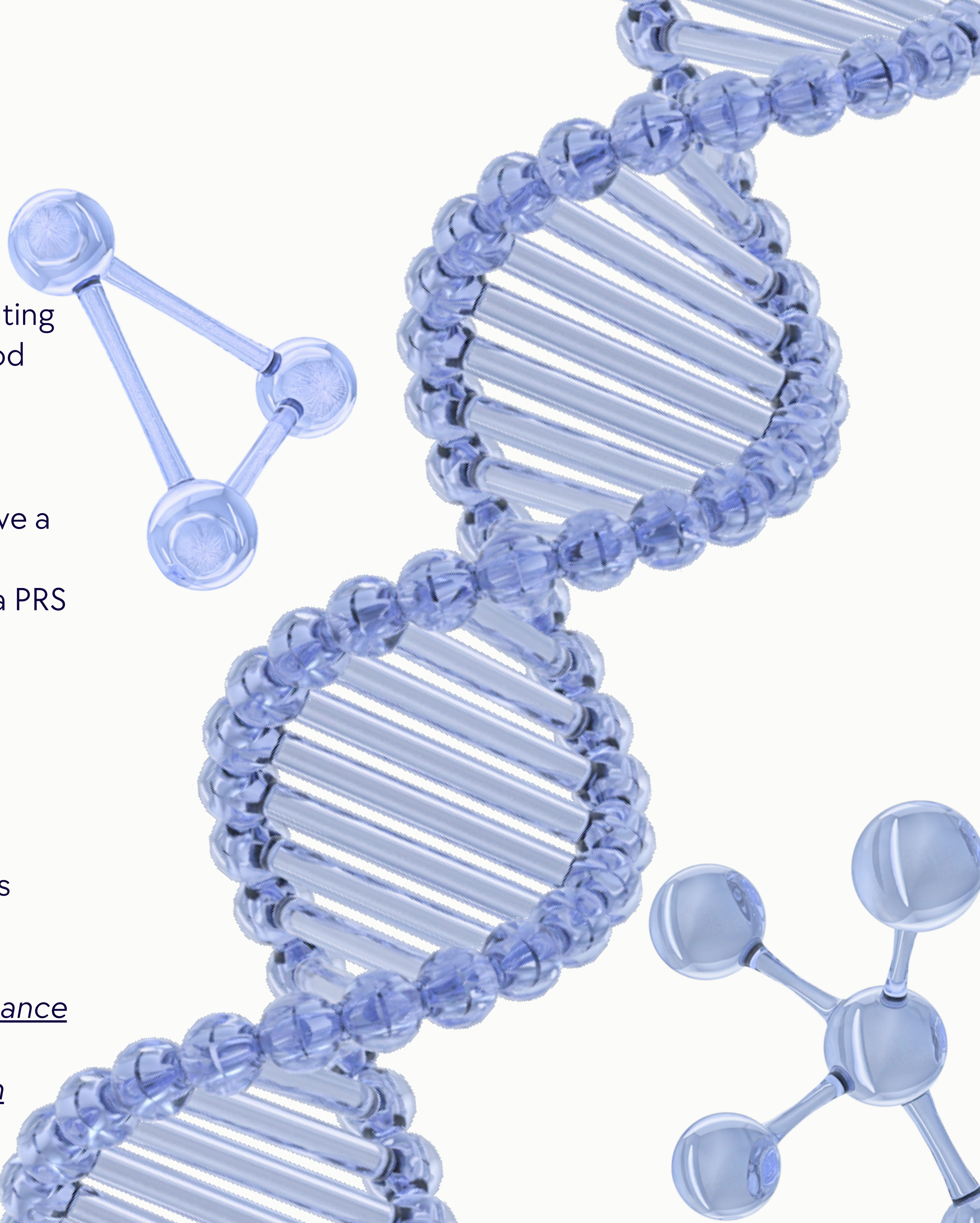
AUC

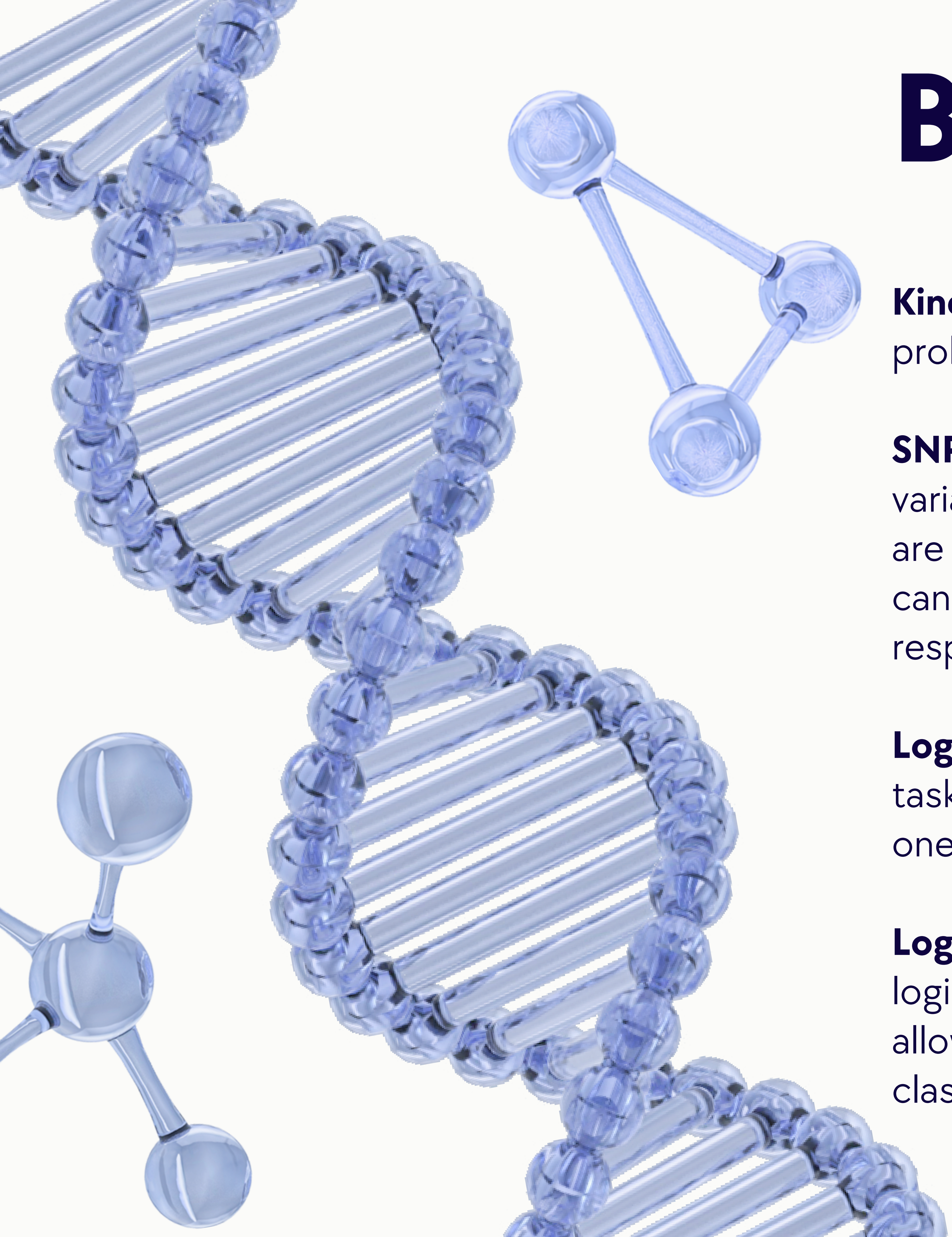
The AUC is a measure of the PRS's ability to discriminate between individuals who have a disease and those who do not. It ranges from 0.5 (no discrimination) to 1.0 (perfect discrimination). A higher AUC indicates better discrimination, which is important for a PRS that is used to predict disease risk.

F1 score

The F1 score is a weighted average of the precision and recall of the PRS. Precision measures the proportion of true positives among all positive predictions, while recall measures the proportion of true positives among all actual positives. The F1 score balances these two measures by giving equal weight to both precision and recall. This makes it a useful metric for evaluating the overall accuracy of the PRS.

In summary, both the AUC and F1 score are useful metrics for evaluating the performance of a PRS, because they provide information about its ability to discriminate between individuals who have a disease and those who do not, as well as its overall accuracy in predicting disease risk.





Brief explanations

Kinetic energy it is described simply as the sum of squared probabilities.

SNPs (Single Nucleotide Polymorphisms): These are tiny genetic variations that involve a single DNA building block, or nucleotide, and are the most common type of genetic variation among people. They can influence traits, susceptibility to diseases, and how individuals respond to drugs and environmental factors.

Logistic Regression: A statistical model used for binary classification tasks, which predicts the probability of an event occurring based on one or more predictor variables.

Logistic Regression with Weights: A statistical model similar to logistic regression, but with the inclusion of weighted data points, allowing for differential emphasis on specific observations during classification.

Data and Methodology

- Our experimental dataset, sourced from UK BioBank, includes approximately 250,000 cases, each corresponding to a female's breast cancer status (either present or absent). It's essential to emphasize that this dataset is highly imbalanced, with only 5% of cases indicating a positive diagnosis of breast cancer. For each person we have taken in consideration around 6000 SNPs.

- For feature engineering we have use kinetic energy also called informational energy.

- We divided the data in 4 intervals of kinetic energies and for evaluation we did 10 cross validation with a 67% train models and 33% test on which we trained and analysed AUC and F1 score of Logistic Regression model.

- Considering the statistical model exposed below S=$\{p\varepsilon = p(x; \varepsilon) I\varepsilon = (\varepsilon 1, , , , , \varepsilon n) \in E\}$ (1)The informational energy on S is a function I:E$\rightarrow R$ defined by,I(ε)=$\int px$ 2(x,ε)dx.

Here are the steps outlined in a more concise manner

1. We determine the kinetic energy for a set of 6000 SNPs.

2. We divide the kinetic energies into 4 intervals.

3. We form groups containing SNPs corresponding to each interval.

4. We train a Logistic Regression model (with weights) for each group from step 3.

Gallery

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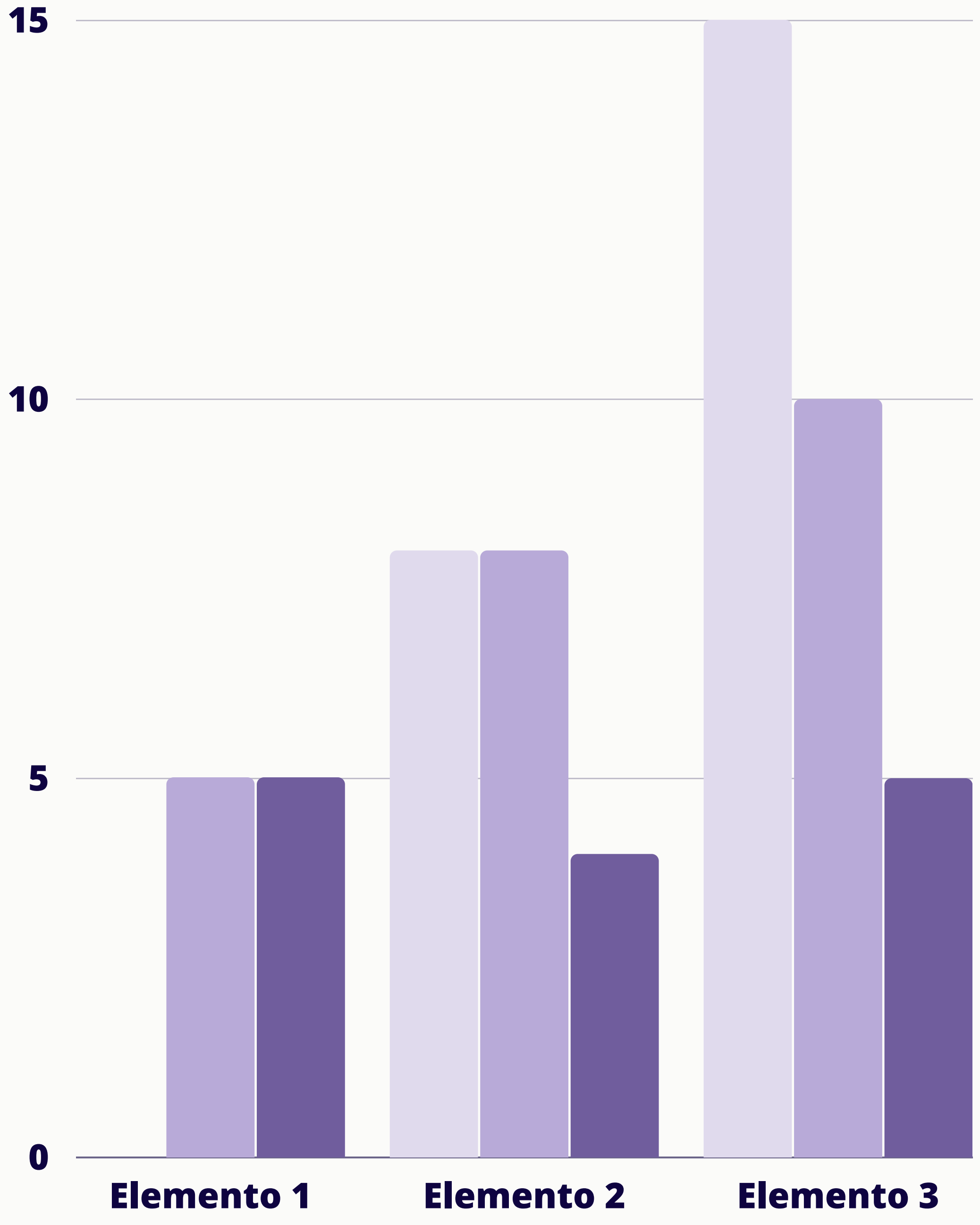

Data Analysis

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Conclusions

- ***Conclusion***

Our findings suggest that by organizing the data using an engineered feature (kinetic energy), we achieved significantly improved results compared to our numerous previous attempts.
Our goal is to explore other variations of this approach and, if feasible, strive for continuous enhancements.

- ***Results***

Based on our analysis across 10 cross-validation iterations, we have observed approximate results as follows:

F1 score

AUC:

THANK YOU